

Motion-consistent 4D cardiac reconstruction and segmentation on public CMR proxies: a methods extension of ReconSeg3D

Working manuscript draft (methods paper)

Status: Publication-oriented methods scaffold. Quantitative panels use **measured demo/smoke metrics only**.

Clinical subject-level public-benchmark tables remain **待补充**.

Target genre: Nature-family / MedIA-style methods · English

Code: <https://github.com/Coucou2016/ReconSeg3D-cardiac-rec>

Abstract

Sparse short-axis (SA) cine cardiac magnetic resonance (CMR) stacks undersample three-dimensional anatomy across the cardiac cycle, motivating dense spatiotemporal reconstruction before structure-aware analysis. Gao et al. (*npj Digital Medicine*, 2026) showed that reconstructed bi-ventricular cine volumes can support long-horizon major adverse cardiovascular event (MACE) prediction on a large **private** acute myocardial infarction (AMI) cohort; that clinical discrimination figure (5-year time-dependent AUC 0.934) is not reproducible without the original data and training recipe and is **not claimed here**. We present a **public, motion-consistent 4D methods extension** of a lightweight ReconSeg3D-style codebase: per-frame volumetric decoding, differentiable warp regularization, and an **image-cycle** consistency loss (forward/backward warps of image intensities—not coordinate-composed inverse-consistent flow), jointly trained with multi-structure segmentation and optional phenotype or risk heads. HeartTTable-style fusion is retained as **HeartTTable-lite** (a single class token attending to concatenated spatial/temporal/table keys) and used only as a fusion ablation. On local auto-fake public-proxy smoke runs we verify end-to-end logging of reconstruction (PSNR/MAE), segmentation Dice, and motion proxies (warp L1, image-cycle error, ejection-fraction proxy). Mounted trees under `data/acdc`, `data/mmwhs`, and `data/emidec` are **demo-scale fake NIfTI** (not challenge downloads); real ACDC / MM-WHS / EMIDEC subject-level tables remain **待补充** pending licensed mounts. We do **not** claim private-AMI MACE parity.

Keywords: cardiac MRI; 4D reconstruction; motion consistency; multi-task learning; public benchmarks; methods reproducibility

1. Introduction

Cardiac cine magnetic resonance imaging (CMR) is typically acquired as a stack of two-dimensional short-axis (SA) slices. Slice gaps, through-plane motion, and temporal undersampling leave a sparse observation of bi-ventricular anatomy. Dense three-dimensional (3D) reconstruction across the cardiac cycle—effectively a four-dimensional (4D) volume—can improve anatomical continuity for segmentation and downstream phenotyping.

Gao et al. (*npj Digital Medicine*, 2026) introduced ReconSeg3D to reconstruct temporally resolved bi-ventricular volumes from SA stacks and HeartTTable to fuse imaging with tabular clinical variables for 5-year MACE after percutaneous

coronary intervention in AMI^[1]. That study’s headline discrimination metrics are tied to a private cohort of thousands of patients and are therefore **out of scope** for any local reimplementations that lack those data.

Gap. Lightweight public reimplementations often broadcast a single reconstructed volume across time or omit explicit motion constraints, which weakens claims about “spatiotemporal” fidelity even when reconstruction and segmentation heads are present. Concurrent lines of work emphasize whole-sequence continuity for 4D segmentation^[5] or joint recon–motion–seg unrolling^[3], but a compact, auditable ReconSeg3D-style training contract with named image-cycle regularization and an honest multimodal ablation boundary remains useful for reproducible methods research.

Contribution. We provide and evaluate a motion-consistent 4D training stack on public proxies:

1. Per-frame 3D reconstruction with temporal mixing (default path; broadcast baseline retained for ablation).
2. MotionNet warp losses plus image-cycle consistency, with optional LV/RV volume-curve smoothing and an ejection-fraction (EF) proxy.
3. Compact segmentation and phenotype/risk heads suitable for ACDC-style proxies.
4. HeartTTable-lite as a fusion ablation only—**not** a claim of pairwise three-modality class-token cross-attention parity.
5. An honest claim boundary, demo-data inventory, and smoke ablation matrix for reconstruction, segmentation, and motion metrics.

2. Related work

Reconstruction. Deep learning has become central to accelerated and robust MRI reconstruction^[2]. Cardiac applications increasingly couple reconstruction with motion or segmentation rather than treating them in isolation.

Joint recon–motion–seg. Unified frameworks that estimate groupwise motion while refining reconstruction and segmentation demonstrate benefits of inter-task dependency^[3]. Decoupled motion–shape models address sparse-slice 4D myocardium reconstruction^[4]. Memory-based networks improve whole-sequence 4D cine segmentation by exploiting spatiotemporal continuity^[5].

Multimodal survival. Neural Cox models such as DeepSurv generalize linear proportional-hazards risk scores^[6]. HeartTTable combines spatiotemporal imaging tokens with tabular variables via transformer cross-attention on private AMI data^[1]. Our public codebase exposes a Cox loss API and a lite fusion module but does not reproduce that clinical study.

Positioning. Relative to Gao et al.^[1], this draft is a **methods extension** focused on motion-consistent training and public-proxy evaluation, not a clinical outcome paper. Relative to Qian et al.^[3] and Ye et al.^[5], we do not claim superior Dice on challenge leaderboards; we claim a transparent lightweight contract (per-frame recon + warp + image-cycle + lite fusion ablation) with reproducible smoke logs.

3. Methods

Implementation modules live under ``reconseg3d/models/`` in the public repository (reconstructor, ``motion.py``, ``heart_ttable.py``, ``losses.py``).

3.1 Problem setup

Let $(x \in \mathbb{R}^{B \times C \times T \times D \times H \times W})$ denote a cine volume batch (channels (C) , frames (T) , depth (D) , height (H) , width (W)). Sparse SA sampling is simulated by zeroing unobserved slices. The model predicts a dense reconstruction $(\hat{x})$, optional flow fields for warping consecutive frames, multi-class segmentations (LV, RV, myocardium; optional scar), and task heads for phenotype classification, binary MACE (synthetic), or Cox partial likelihood (when event times are available).

3.2 Per-frame reconstruction

Unlike broadcasting a single decoded volume across (T) , the default path encodes each frame with a compact 3D CNN (optional transformer bottleneck) and decodes per-frame volumes. Temporal mixing couples neighboring frames without collapsing the time axis. Smoke configs typically use spatial grids such as $(D \times H \times W = 16 \times 32 \times 32)$ (demo ACDC volumes on disk are $(16 \times 32 \times 32)$ with $(T=8)$).

3.3 Motion consistency

A MotionNet predicts voxel displacements $((d_z, d_y, d_x))$. Displacements are scaled to grid coordinates with dimension guards for singleton axes. We regularize:

- **Warp L1** between warped source and target intensities.
- **Image-cycle** error: compose forward and backward warps on **images** and penalize deviation from the original. This is **distinct** from inverse-consistent flow-field composition in classical registration; we name it image-cycle to avoid reviewer misreading.
- Optional **volume-curve** smoothness on fractional LV/RV volumes and an **EF proxy**.

3.4 Segmentation and task heads

A compact 3D UNet-style head predicts multi-class masks. Phenotype (ACDC-style), BCE MACE, or Cox risk can be enabled via configuration. Metrics pool at epoch level; tiny fake splits can yield unstable AUC / C-index and must not be over-interpreted.

3.5 HeartTTable-lite (ablation)

When ``fusion: heart_ttable``, a single CLS token attends over concatenated spatial, temporal, and table key-value streams (``HeartTTable.risk_logits``). This is **not** the original three class-token pairwise cross-attention design and is reported only as a fusion ablation against concat-MLP fusion.

3.6 Training protocol (smoke)

Configs under ``configs/`` set compact grids and 1–2 epoch smoke runs. Public loaders may auto-synthesize fake NIfTI when roots are empty (``data.auto_fake``). Subject-level cross-validation, physical-space HD95, and calibration plots are specified for future real-data tables (**待补充**).

4. Experiments

4.1 Datasets and honesty inventory (2026-08-16)

Dataset	Role	On-disk status in this workspace	
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Synthetic / auto-fake Pipeline verification Used for all quantitative panels below			

| ACDC (challenge) | Recon/seg/phenotype proxy | ****Not mounted.**** Local `data/acdc/` is demo fake: 8 patients, 4D ≈ 0.49 MB, shape $\backslash((32,32,16,8)\backslash)$. Official download requires CREATIS registration — ****blocked without credentials**** |

| MM-WHS | Static seg / table fusion | Local `data/mmwhs/` = 4 demo pairs; motion off / $\backslash(T\{=\}1\backslash)$ recommended when real data arrive |

| EMIDEC | Scar / infarct proxy | Local `data/emidec/` = 4 demo pairs; ****待补充**** |

| Private AMI | 5-year MACE | Out of scope (`data/ami/` holds example manifest only) |

4.2 Ablations

Broadcast baseline; per-frame without motion weights; per-frame with warp/cycle; concat vs HeartTTable-lite; phenotype task. Summaries written by `scripts/summarize\_runs.py` from `outputs/ablations\_smoke\_v2/`.

4.3 Metrics

Reconstruction: PSNR, global-SSIM proxy, MAE. Segmentation: per-class Dice; optional subsampled HD95 (approximate on smoke grids). Motion: warp L1, image-cycle error, EF proxy. Phenotype/MACE/C-index when applicable—****demo values are not clinical****.

5. Results

5.1 Smoke ablation summary (****DEMO ONLY**** — measured)

Quantitative panels in Figures 2–4 are generated with SciencePlots from `outputs/ablations\_smoke\_v2/table.csv` (measured local smoke, not fabricated). Representative reconstruction PSNR on motion-relevant runs is ≈ 19.1 dB with MAE ≈ 0.69 under the compact fake-data regime. Motion runs log non-zero warp and image-cycle errors. Segmentation Dice on fake NIfTI is low and class-imbalanced (RV often near zero)—consistent with under-trained smoke rather than a clinical ceiling.

****Table 1. Reconstruction / motion smoke (DEMO).**** Selected rows from `ablations\_smoke\_v2/table.csv`.

Run	PSNR (dB)	MAE	Warp L1	Image-cycle	EF proxy
broadcast_baseline	19.08	0.693	—	—	—
per_frame_no_motion	19.15	0.692	$9.7\text{e-}4$	≈ 0	0.088
per_frame_motion	19.15	0.695	$1.1\text{e-}3$	$3.2\text{e-}4$	0.081
fusion_concat	19.09	0.692	$5.0\text{e-}4$	$1.1\text{e-}4$	0.146
fusion_heart_ttable	19.26	0.689	$1.3\text{e-}3$	$7.6\text{e-}4$	0.068

****Table 2. Segmentation Dice smoke (DEMO).****

Run	Dice LV	Dice RV	Dice MYO	Dice mean
broadcast_baseline	0.227	≈ 0	0.278	0.168
per_frame_no_motion	0.014	≈ 0	0.495	0.170
per_frame_motion	0.038	≈ 0	0.439	0.159
fusion_heart_ttable	≈ 0	≈ 0	0.567	0.189
task_phenotype	0.053	≈ 0	≈ 0	0.018

Phenotype smoke: logged phenotype accuracy 0.0 with AUC 1.0 on a tiny fake split illustrates metric instability; we treat this as a **pipeline check**, not a result. Columns labeled MACE AUC / C-index in smoke CSVs are likewise synthetic-label diagnostics.

Paper recon / ACDC smoke checkpoints (`outputs/paper_*_smoke/metrics.json`) similarly show low Dice and mid-teens–19 dB PSNR after one epoch.

5.2 Figures

- **Fig. 1** Pipeline schematic (roles of recon, motion, seg, lite fusion).
- **Fig. 2** Reconstruction PSNR/MAE ablation (**DEMO**).
- **Fig. 3** Warp / image-cycle / EF proxies (**DEMO**).
- **Fig. 4** Dice heatmap across ablations (**DEMO**).
- **Fig. 5** Claim-boundary diagram vs original clinical paper.

Real ACDC cine volume curves, confidence intervals, and physical HD95: **待补充**.

6. Discussion

The practical contribution of this codebase is a **reproducible motion-consistent 4D training contract** on public proxies, with documentation that prevents silent overclaim of the original AMI discrimination numbers. Image-cycle naming matters for peer review: reviewers familiar with registration literature will expect inverse consistency of flow fields; we instead regularize intensity cycles after warping.

HeartTTable-lite is intentionally under-powered relative to the original multimodal module. Building full three-CLS pairwise attention before real AMI data would optimize for a claim we cannot evaluate publicly.

Relative to whole-sequence memory segmentation^[5] and unrolled joint recon–motion–seg^[3], our novelty claim is scoped to an auditable lightweight ReconSeg3D extension with explicit motion losses and fusion ablation boundaries—not leaderboard dominance.

Limitations. Compact backbones; simulated sparse SA (not vendor k-space); approximate HD95/SSIM; no nested CV; no private AMI; on-disk “ACDC/MM-WHS/EMIDEC” trees are demo fakes until challenge data are licensed and mounted; Chinese/English bilingual materials in the companion research report for local review, while this draft remains English for Nature-family methods style.

7. Conclusions

We describe a methods-oriented, motion-consistent 4D extension of a ReconSeg3D-style pipeline for public cardiac MRI proxies, with HeartTTable-lite as a fusion ablation and an explicit refusal to claim private AMI AUC 0.934. Measured smoke ablations demonstrate that reconstruction, segmentation, and motion metrics are logged end-to-end. Completing subject-level public-data tables after licensed dataset mounts and—only with authorized private cohorts—survival evaluation remain future work.

Data availability

Public benchmarks (ACDC, MM-WHS, EMIDEC) follow their licenses (`docs/DATA.md`). Official ACDC access requires CREATIS registration; this repository does not redistribute challenge volumes. Smoke metrics and SciencePlots figure sources live under `outputs/` (local) and `artifacts/paper/figures/`. No private AMI data are distributed with this repository. Demo trees under `data/` are synthetic placeholders for CI and pipeline checks.

Code availability

<https://github.com/Coucou2016/ReconSeg3D-cardiac-rec>

Install via `requirements.txt` (includes SciencePlots for figure regeneration). Reproduce smoke figures with `python scripts/make\_paper\_figures.py` and the research report with `python scripts/build\_report\_bundle.py`.

References

[^1]: Gao, Q. et al. 3D Spatiotemporal cardiac reconstruction for predicting MACE in acute myocardial infarction. *npj Digit. Med.* **9**, 325 (2026). <https://doi.org/10.1038/s41746-026-02449-0>

[^2]: Hammernik, K. et al. Deep learning for accelerated and robust MRI reconstruction. *MAGMA* (2024). <https://doi.org/10.1007/s10334-024-01173-8>

[^3]: Qian, P. et al. Unified Deep Learning for Simultaneous Cardiac Cine MRI Reconstruction, Motion Estimation and Segmentation. *IEEE ISBI* (2024). <https://doi.org/10.1109/ISBI56570.2024.10635390>

[^4]: Yuan, X. et al. 4D Myocardium Reconstruction with Decoupled Motion and Shape Model. *ICCV* (2023).

[^5]: Ye, M., Xin, B., Axel, L. & Metaxas, D. Continuous Spatio-Temporal Memory Networks for 4D Cardiac Cine MRI Segmentation. In *WACV* 9514–9524 (2025). Also [arXiv:2410.23191](https://arxiv.org/abs/2410.23191).

[^6]: Katzman, J. L. et al. DeepSurv: personalized treatment recommender system using a Cox proportional hazards deep neural network. *BMC Med. Res. Methodol.* **18**, 24 (2018). <https://doi.org/10.1186/s12874-018-0482-1>

Notes (nature-writing)

Axes: `task=manuscript` · `paper\_type=methods` · `language=en` · `journal=nature-family` (methods framing).

Claim-evidence map

Claim	Evidence	Status
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Motion-consistent 4D stack exists and logs metrics	Code + measured smoke tables + Figs 2–3	Supported (demo regime)
Public clinical superiority vs baselines	Real ACDC folds	Needs evidence (**待补充**)
HeartTTable parity / AMI AUC 0.934	—	Out of scope / rejected

Assumptions or missing inputs: licensed ACDC/MM-WHS/EMIDEC on disk; physical HD95; nested CV; private AMI for survival claims.

Why this structure: methods argument chain (problem → method → ablation → boundary); clinical multimodal paper deferred.

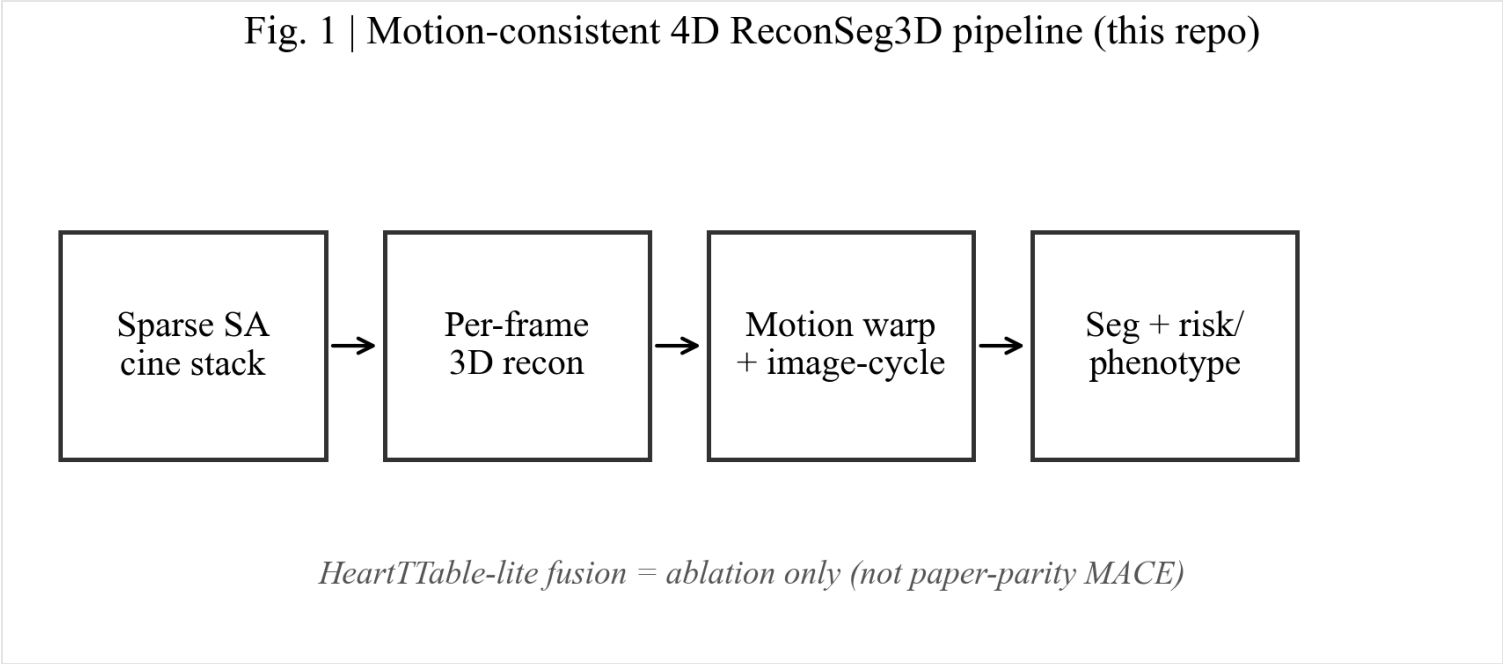


Figure 1
Pipeline schematic (SciencePlots).

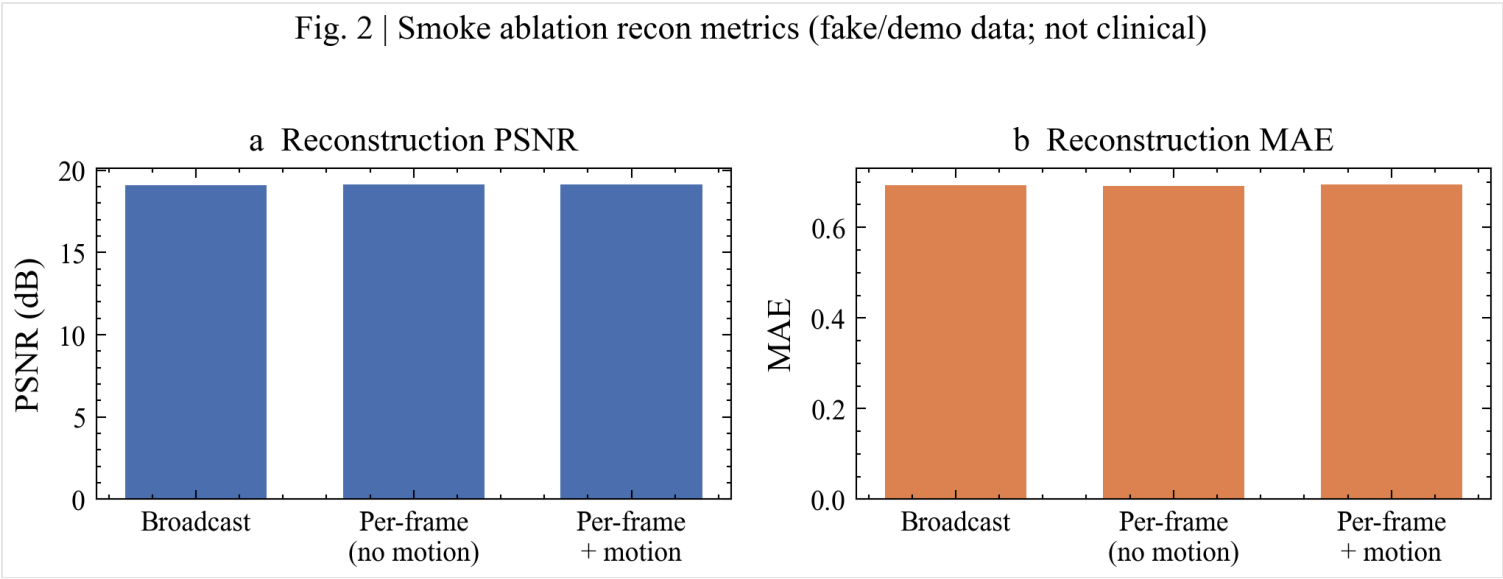


Figure 2
Reconstruction smoke ablation.

Fig. 3 | Motion / volume-curve proxies on smoke ablations (demo)

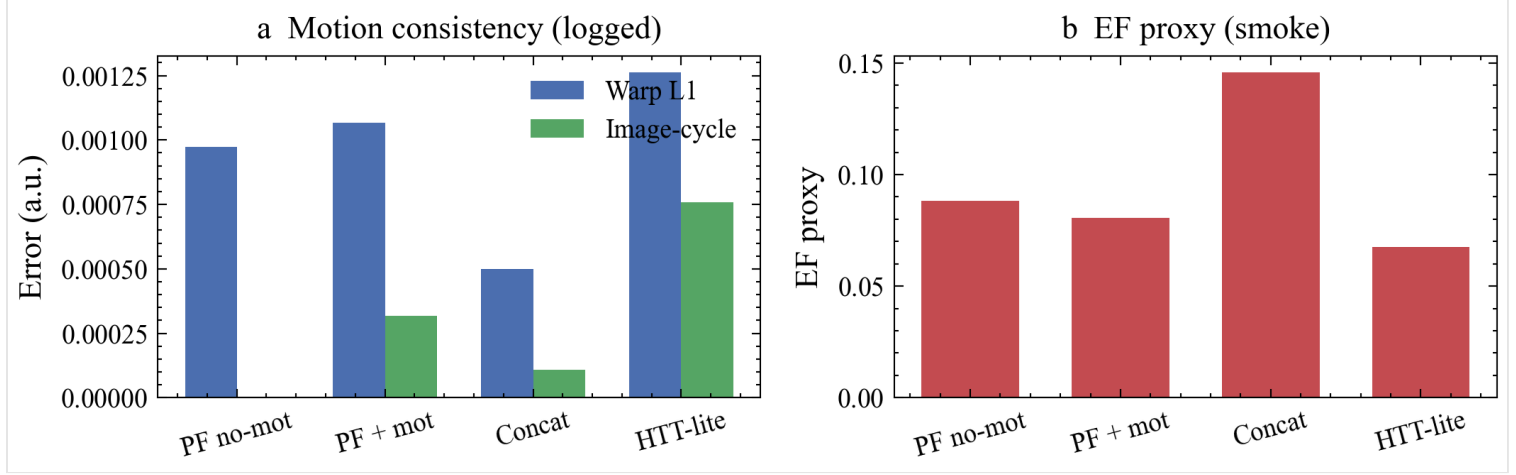


Figure 3
Motion proxies (demo).

Fig. 4 | Segmentation Dice heatmap (smoke / fake NIfTI; not clinical)

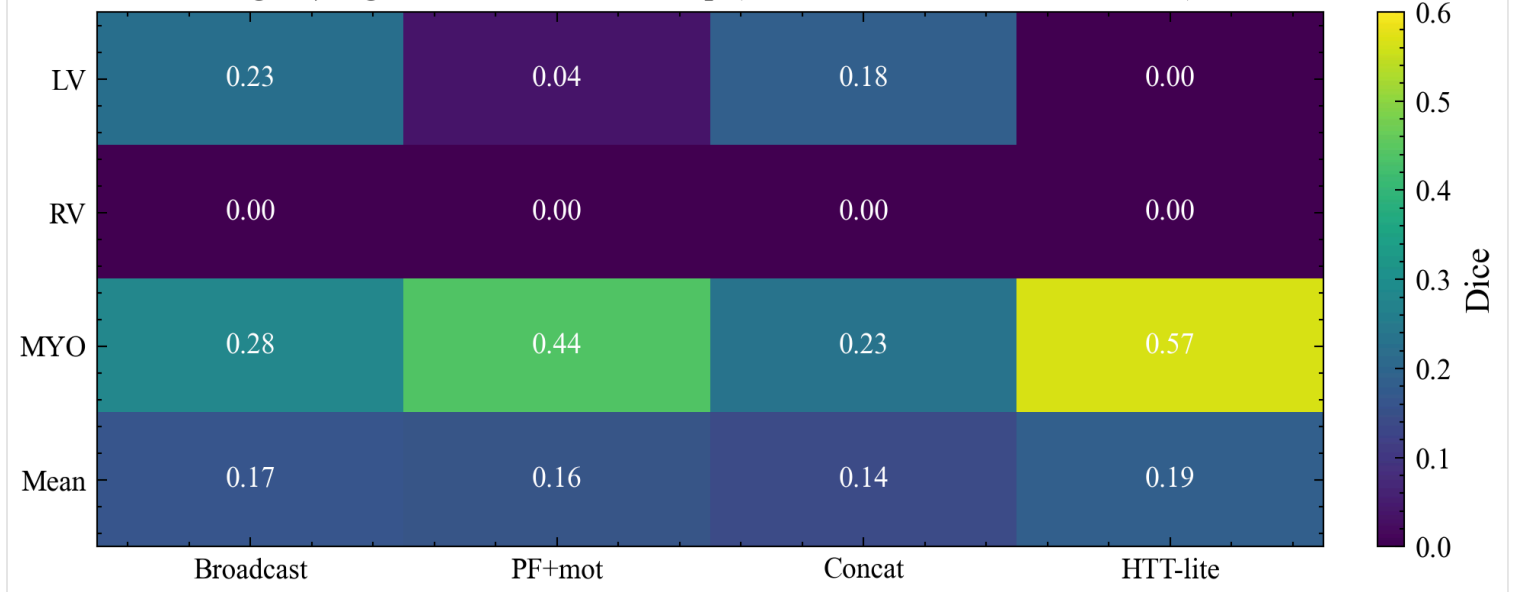


Figure 4
Dice heatmap (demo).

Fig. 5 | Innovation claim boundary vs original npj Digit. Med. paper

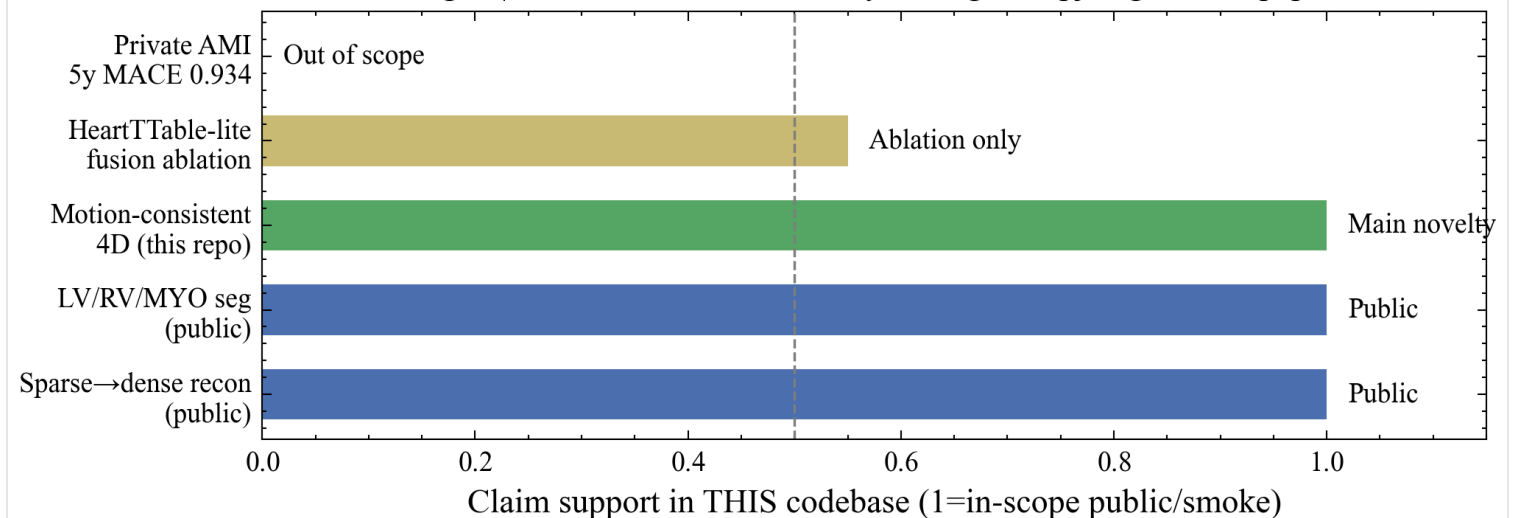


Figure 5
Claim boundary vs original paper.